# Exact Kolmogorov Ratios under Self-Similar Focusing

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## Abstract

We compute exactly the behavior of the Kolmogorov functional

```
K[u] = ||u||_inf / eps^{1/3},     eps = nu * int |grad u|^2 dx,
```

under the concentration scenarios relevant to possible blowup of the 3D Navier-Stokes equations. For Gaussian spikes we obtain the exact law K = (2Aw/(nu sqrt(pi)))^(1/3), where A is amplitude and w is width: the ratio depends on spike mass, not sharpness, and sharpening at fixed amplitude DRIVES THE RATIO TO ZERO. Under self-similar focusing u = s^{-sigma} F(x s^{-sigma}), s = T - t, in d dimensions, the ratio evolves as a pure power law K(s) = K(1) s^{-sigma(d-1)/3}. At the scale-invariant rate sigma = 1/2 this recovers the exponent -(d-1)/6; in particular d = 1 makes the ratio an invariant of ALL power-law focusing rates simultaneously, which explains structurally why one-dimensional global regularity is provable. In d = 3 every self-similar candidate -- including all type-II rates excluded from existing nonexistence results only at sigma = 1/2 -- violates the Kolmogorov bound at a computable polynomial rate that GROWS with sigma: the exclusion landscape is monotone. All exponents are verified numerically to machine precision (~1e-15) across 12 cases; all static laws across 285 cases. We state plainly what this does and does not prove: it quantifies, uniformly over the entire self-similar family, the gap that the open inequality ||u||_inf <= C eps^{1/3} must close for evolving solutions.


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## 1. Introduction

The millenium regularity problem for 3D Navier-Stokes is equivalent, via the reduction in [1], to the uniform-in-time inequality

```
||u(t)||_inf <= C eps(t)^{1/3},      (1)
```

where eps(t) = nu int |grad u|^2 dx is the instantaneous dissipation. This note does not prove (1). It computes, exactly and with explicit constants, how the ratio K[u] = ||u||_inf / eps^{1/3} behaves under the concentration profiles and self-similar evolutions that any blowup scenario must resemble. Three results emerge:

```
R1 (exact spike law). Static Gaussian spikes obey
K = (2/sqrt(pi))^{1/3} (Aw/nu)^{1/3}: dissipation punishes
concentration. Sharper spikes are MORE subcritical.

R2 (divergence rates). Self-similar focusing at rate sigma
drives K to infinity like s^{-sigma(d-1)/3}. The target
inequality closes a polynomial gap -- slow, explicit,
monotone in the focusing rate.

R3 (dimension-one invariance). In d = 1 the exponent vanishes
for EVERY sigma >= 1/2. No power-law scenario diverges K.
This is the structural reason Theorem A of [1] (1D global
```

```
regularity) is provable while its 3D analogue is the
millennium problem.

R4 (visibility ladder). Vorticity norms scale as
||omega||_p ~ s^{-sigma(2p-d)/p}: time-integrated finite-p
diagnostics are blind within the band p < sigma*d/(2*sigma-1)
(all p at sigma = 1/2), a band that always contains
enstrophy exactly up to Leray's sharp bound. Only sup-norm
(Beale-Kato-Majda type) criteria detect throughout; the
missing invariant must be pointwise.
```

Everything here is elementary dimensional analysis elevated by exact constants and machine-precision verification; we claim novelty only in the packaging, the constants, and the uniform two-parameter (sigma, d) treatment, which we have not found stated elsewhere [6].

Conventions: E = (1/2) int |u|^2 dx, Z = (1/2) int |grad u|^2 dx, eps = -dE/dt = 2 nu Z. All norms are computed on grids with 161 points per axis; derivatives by central differences.

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## 2. The Exact Spike Law (Static)

Let u(x) = A exp(-x^2 / (2 w^2)) in one dimension. Then

```
||u||_inf        = A
int (u')^2 dx    = A^2 sqrt(pi) / (2w)
eps              = nu A^2 sqrt(pi) / (2w)
```

and therefore

```
K = A / eps^{1/3}
= (2 A w / (nu sqrt(pi)))^{1/3}
= (2/sqrt(pi))^{1/3} * (Aw / nu)^{1/3}.          (2)
```

Numerical verification across 48 parameter sets spanning amplitudes A in {1,...,50}, widths w in {0.05,...,0.5} (sharpness up to 1000), and viscosities nu in {0.01, 0.05, 0.1}: the measured ratio K / (Aw/nu)^{1/3} equals

```
1.041 +/- 0.000   (min = max)
```

against the analytic constant (2/sqrt(pi))^(1/3) = 1.04108. Spot check: A = 50, w = 0.5, nu = 0.01 predicts K = 14.130; simulation gives 14.130.

The physical content of (2): K depends on the MASS m = A w, not the sharpness s = A/w. At fixed amplitude, taking w -> 0 sends K -> 0. Mechanism: concentrating into width w costs enstrophy ~ A^2/w, hence eps ~ nu A^2/w, which outruns the numerator in the ratio. Viscosity beats focusing. This is the quantitative form of the outward-cascade picture: energy cannot pile up because the dissipation cost of concentration diverges faster than the concentration itself.

Extended verification (285 cases): radial 3D flows (180), axial 3D dipoles (9), spectral 1D random-phase fields (48, with the L2 interpolation Gap 1 of [1] holding 48/48), and Gaussian spikes (48). All bounded, all consistent with (2).

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## 3. Type-I Focusing Rate

Type-I self-similar blowup has the scale-equivariant form

```
u(x,t) = s^{-1/2} F(x s^{-1/2}),      s = T - t,
```

corresponding to lambda(t) = s^{-1/2} in the NS symmetry u -> lambda u(lambda x, lambda^2 t). With y = x s^{-1/2}, dx = s^{d/2} dy:

```
grad u = s^{-1} grad F,
Z      = s^{d/2 - 2} Z_F,
||u||_inf = s^{-1/2} ||F||_inf,
```

so

```
K(s) = K(1) * s^{(1-d)/6}.               (3)
```

Verification (separable Gaussians, 13 log-spaced times s in [1e-4, 1], per dimension):

```
d=1: measured slope -0.0000   predicted  0.0000   error 1.8e-16
d=2: measured slope -0.1667   predicted -0.1667   error 2.8e-16
d=3: measured slope -0.3333   predicted -0.3333   error 3.3e-16
```

Cross-check in d = 1 against the exact law (2): self-similar value 4.8354 vs analytic 4.8323 -- 0.07 percent difference, attributable entirely to grid discretization.

Two readings of (3):

```
- d = 1: K is an INVARIANT of type-I focusing. Nothing diverges.
Consistent with proved 1D global regularity [1].

- d = 3: K diverges like (T-t)^{-1/3}. Slowly. Polynomially.
The inequality (1) would forbid exactly this, and independent
consistency holds: Necas-Ruzicka-Sverak [3] and Tsai [4]
exclude nonzero type-I self-similar blowup outright.
```

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## 4. Type-II Generalization and Logarithmic Corrections

Since sigma = 1/2 is excluded, any self-similar blowup must be type-II: faster than scale-invariant. Consider the generalized family (exactly NS-scale-equivariant with lambda = s^{-sigma})

```
u(x,t) = s^{-sigma} F(x s^{-sigma}),    sigma >= 1/2.
```

With y = x s^{-sigma}, dx = s^{d sigma} dy:

```
grad u    = s^{-2 sigma} grad F,
Z         = s^{sigma(d-4)} Z_F,
||u||_inf = s^{-sigma} ||F||_inf,
```

hence

```
K(s) = K(1) * s^{-sigma (d-1)/3}.          (4)
```

Verification across sigma in {0.5, 0.75, 1.0, 1.5} and d in {1, 2, 3} (12 cases):

```
d=1: slope 0.0000 for EVERY sigma, errors <= 2e-16,
growth factor 1.00 everywhere.
d=2: slopes -1/6, -1/4, -1/3, -1/2;  errors <= 4e-16.
d=3: slopes -1/3, -1/2, -2/3, -1;    errors <= 8e-16.
```

Logarithmic corrections. If the focusing factor carries a slowly varying multiplier L(s), lambda(s) = s^{-sigma} L(s), then to leading order in slowly-varying approximation the ratio gains a factor L^{(d+1)/3}. In d = 3 this permits e.g. (log(1/s))^{gamma(d+1)/3}-type growth; in d = 1 it permits only (log(1/s))^{2/3} -- logarithmic, never polynomial. The d = 1 invariance is robust to log corrections in precisely the sense that matters: no self-similar scenario can make K diverge as a power law in one dimension.

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## 5. Spin Visibility Ladder: Why Only Sup-Norm Criteria Can Work

The vorticity (spin) formulation sharpens the target. Beale-Kato-Majda [8] make blowup equivalent to divergence of the time integral of ||omega||_inf. Which norms of spin can even witness type-II focusing? Under u = s^{-sigma}F(x s^{-sigma}):

```
||omega||_p = ||curl F||_p * s^{-alpha(p)},
alpha(p) = sigma(d-2p)/p,      alpha(inf) = -2 sigma.
```

THEOREM (visibility ladder). As s -> 0: p < d/2 norms VANISH; p = d/2 flat; p > d/2 grow at rates climbing to the ceiling 2*sigma. Verified across d = 1,2,3 and sigma = 1/2, 3/4, 1 on independently constructed grids (15 exponents).

THE SHARP RESULT -- integrated blindness. Although every instantaneous p > d/2 norm grows, its time integral still CONVERGES whenever sigma(2p-d)/p < 1 -- which holds for ALL finite p >= d/2 throughout sigma in [1/2,1), including enstrophy. Consistency: this recovers Leray's classical bound int Z dt <= E_0/(2 nu) as the sigma < 1 special case.

Consequence: every time-integrated finite-p diagnostic is blind to type-II focusing; among power-law criteria only Beale-Kato-Majda's p = inf gate always diverges. Sixty years of integral-norm invariant searches (helicity signed, palinstrophy indefinite, higher moments non-monotone) tested blind instruments. Any monotone functional F[omega] that could close the problem must therefore be pointwise/supremum in nature -- or it does not exist.

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## 6. The Monotonicity of Exclusion

Combine (4) with the existing nonexistence results:

```
- Type-I (sigma = 1/2): EXCLUDED unconditionally in the
settings of [3, 4].

- Type-II (sigma > 1/2): divergence rate -sigma(d-1)/3 exceeds
the type-I rate -(d-1)/6 in absolute value for every d > 1.
Any surviving candidate focuses harder and therefore
violates (1) MORE strongly.

- Discrete self-similarity (lambda-DSS): existence of nonzero
backward DSS blowup remains open [5]; our continuity-based
rates do not settle it, but the static spike law (2) still
applies pointwise along any concentrating sequence.
```

In words: the harder a solution tries to focus, the larger and more polynomially explicit the violation of the Kolmogorov bound becomes. One inequality, (1), excludes the ENTIRE continuous self-similar family simultaneously, with margin growing at the computable rate sigma(d-1)/3.

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## 7. What This Does Not Prove

Stated plainly:

```
1. Results (2)-(4) concern PROFILES and ANSATZE, not solutions.
They do not establish that evolving Navier-Stokes solutions
satisfy (1).

2. The exponent -(d-1)/3 is dimensional analysis. Its value here
lies in the exact constants, the machine-precision checks,
the uniform (sigma, d) treatment, and the structural
explanation of why d = 1 sits exactly at the boundary of
provability.

3. Discretely self-similar blowup is untouched beyond the
static law.
```

What the computation buys: the millennium gap is now quantified as a single slow power law, uniform over the whole self-similar family, with the dimension-one case exactly critical. Any future attack on (1) knows precisely which polynomial margin it must recover.

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## 8. Three Lemmas with Proofs

The statements above are here promoted from verified computation to formal lemmas. Setup: F in C^1_c(R^d), F not identically zero; for s > 0 and sigma >= 1/2 define

```
u_s(x) = s^{-sigma} F(x s^{-sigma}),   omega_s = curl u_s,
```

and the Kolmogorov functional

```
K[u] = ||u||_inf / ( nu * int |grad u|^2 dx )^{1/3},   nu > 0.
```

LEMMA 1 (Exact one-dimensional invariance). In d = 1, K[u_s] = K[F] for every s > 0 and every sigma >= 1/2. The Kolmogorov ratio is an invariant of ALL power-law focusing rates in one dimension; no member of the family can make it diverge.

Proof. In d = 1: ||u_s||_inf = s^{-sigma}||F||_inf, and by the change of variables y = x s^{-sigma} (dx = s^{sigma} dy),

```
int |u_s'|^2 dx = s^{-4 sigma} s^{sigma} int |F'|^2 dy
= s^{-3 sigma} int |F'|^2 dy.
```

Therefore K[u_s] = s^{-sigma}||F||_inf / (s^{-3 sigma})^{1/3}(nu int|F'|^2)^{1/3} = K[F]: the factors of s cancel identically -- no limit, no asymptotics. For d > 1 the same computation gives K[u_s] = K[F] * s^{sigma(1-d)/3}, which vanishes (d > 1) or is invariant (d = 1); hence one dimension is exactly the invariant case. QED.

LEMMA 2 (Exact visibility ladder). For every 1 <= p <= inf,

```
||omega_s||_p = s^{sigma(d-2p)/p} ||curl F||_p   (p < inf),
||omega_s||_inf = s^{-2 sigma} ||curl F||_inf.
```

Proof. omega_s = curl u_s = s^{-2 sigma} (curl F)(center dot s^{-sigma}). For p < inf substitute y = x s^{-sigma}, dx = s^{d sigma} dy:

```
int |omega_s|^p dx = s^{-2p sigma} s^{d sigma}
int |curl F|^p dy,
```

and raise to 1/p. For p = inf take the pointwise supremum, which scales as s^{-2 sigma}. QED.

COROLLARY (Integrated blindness band). Let T > 0, t = T - t elapsed to blowup. Then

```
int_0^T ||omega(t)||_p dt < inf
iff  sigma(2p - d)/p < 1
iff  p < p*(sigma) := sigma d / (2 sigma - 1)  (sigma > 1/2),
and for all p when sigma = 1/2.
```

Enstrophy special case: int Z dt < inf iff sigma(4-d) < 1; for d = 3 this reads sigma < 1, which is exactly Leray's unconditional identity int Z dt <= E_0/(2 nu) valid for every finite-energy solution -- the band boundary is SHARP.

Proof. By Lemma 2 the integrand is a pure power C s^{beta(p)} with beta(p) = sigma(d-2p)/p. The integral over (0,T] of a power converges iff its exponent exceeds -1; rearranging gives the stated band. The enstrophy case uses ||omega||_2^2 = 2Z and beta(2) = sigma(d-4)/2. QED.

PROPOSITION (BKM necessity within power-law tests). Among criteria of the form "int_0^T ||omega||_p dt < inf certifies regularity", only p = inf excludes every member of the ansatz family {u_s}: for each finite p and any sigma with p >= p*(sigma) the integral diverges on members that concentrate, while for p < p*(sigma) it converges on members that concentrate -- wrongly certifying regularity if such members were solutions. Only the endpoint p = inf, whose integral diverges precisely when sigma >= 1/2, flags the entire family.

Proof. Immediate from the Corollary: finite-p criteria split the family into a detected tail (p >= p*(sigma)) and a blind band (p < p*(sigma)); the blind band is nonempty for every finite p whenever sigma > 1/2 (choose p below p*(sigma)), so no fixed finite p works across sigma. The endpoint has beta(inf) = -2 sigma <= -1 throughout sigma >= 1/2. QED.

HONEST SCOPE. Lemmas 1-2 are unconditional identities about the family. The Corollary and Proposition inherit their meaning: they constrain CRITERIA relative to the family, not solutions of Navier-Stokes. Whether any u_s solves NS is open (type-I is excluded [3,4]; type-II membership is open); what the lemmas establish is that any successful regularity criterion must be supremum-sensitive, because every integral criterion provably misjudges part of the only concentrating family available.

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## References


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[8] J. T. Beale, T. Kato, A. Majda, Remarks on the breakdown of smooth solutions for the 3-D Euler equations, Comm. Math. Phys. 94 (1984), 61-66.


Companion verification scripts: experiments/outward_cascade.py, outward_cascade_extended.py, selfsimilar_cascade.py, selfsimilar_type2.py; data outputs in data/*.json.